

Real Analysis I

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Contents

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Homework 2: Week 2 Due Sep 9

Mon 29 Aug 2022 14:24

I use the following lemma to prove Exercise 3.J.

Lemma 1 (pigeonhole principle). Let Y be a finite set and X be finite with cardinality greater than $|Y|$. Then for any function $f : X \rightarrow Y$, there exists $x_1, x_2 \in X$ such that $f(x_1) = f(x_2)$.

Proof. Denote A_y by the set $\{x \in X : f(x) = y\}$. We must have $|A_y| < 2$ for all $y \in Y$, otherwise there will be more than one element from X mapped to the same element in Y . Hence $0 \leq |A_y| \leq 1$. We have

$$\left| \bigcup_{y \in Y} A_y \right| = \sum_{y \in Y} |A_y| \leq \sum_{y \in Y} 1 = |Y|.$$

The first equality comes from the fact that A_y are all disjoint and finite (since elements mapped to different targets must be different).

But for any $x \in X$, it is mapped to some element $y = f(x)$. In other words, $x \in A_{f(x)} \subset \bigcup_{y \in Y} A_y$. In the other direction, for any $x \in \bigcup_{y \in Y} A_y$, it is some element in X by definition. We have shown that $X = \bigcup_{y \in Y} A_y$ and therefore $|X| = \left| \bigcup_{y \in Y} A_y \right| \leq |Y|$ which is a contradiction to the assumption that $|X| > |Y|$. Therefore there must exist some y such that $|A_y| \geq 2$, which is exactly what we want to prove. $\square$

I prove this exercise in extra, because I used it in Exercise 7.I.

7.I. Show that the collection of "right hand" end points in $\mathbf{F}$ is denumerable. Show that if all these end points are deleted from $\mathbf{F}$, then what remains can be

put onto one-one correspondence with all of $[0, 1)$. Conclude that the set $\mathbf{F}$ is not countable.

Proof. Each "right hand" end point is in the form $\frac{k+1}{3^n}$ where $k, n \in \mathbb{N}$. Therefore it is a rational number, and the set of rational numbers $\mathbb{Q}$ is denumerable. Because the set of right hand end points is a subset of rational numbers, it is denumerable.

Now we construct cascaded segmentation of the half-open interval $[0, 1)$. For each nonnegative integer n and $0 \leq k \leq n$, define A_{nk} to be

$$A_{nk} = [\frac{k}{2^n}, \frac{k+1}{2^n}).$$

We use induction to verify the following properties:

1. for all n there are exactly 2^n disjoint sets of the form A_{nk} .
2. (disjoint segmentation) for all n and all distinct k, l ,

$$A_{nk} \cap A_{nl} = \emptyset.$$

3. (filling) for all n ,

$$\bigcup_{0 \leq k \leq n} A_{nk} = [0, 1).$$

4. (branch split) for any n, k , $A_{n+1,l} \subset A_{nk}$ if and only if

$$2k \leq l \leq 2k + 1.$$

5. (cascading property) whenever $x \in A_{nk}$ and $x \in A_{ml}$ where $n \leq m$, we have $A_{ml} \subset A_{nk}$.

Now for each n , we can construct a bijection $G : \bigcup_{0 \leq k \leq n} A_{nk} \rightarrow F_n^*$, where F_n is as defined in the textbook, and F_n^* is F_n with right endpoints removed. There are exactly 2^n disjoint half-open intervals A_{nk} . There are also exactly 2^n disjoint half-open intervals in F_n^* . We map the k -th interval A_{nk} to the k -th interval in F_n^* . Note that any two half-open intervals on $\mathbb{R}$ have a bijection between them.

Now we show that there exists a bijection $\mathcal{F} : [0, 1) \rightarrow F^*$ where F^* is the Cantor set with right endpoints removed. Take any $x \in [0, 1)$. For any $n \in \mathbb{N}$, let A_{nk} be the unique half-open interval in the construction that contains x . Now G maps A_{nk} to some half-interval B_n in the construction of F_n^* . Now $B_{n+1} \subset B_n$ and $B_n \neq \emptyset$. Denote b_n as the left endpoint of B_n . Now $y = \sup_{n \in \mathbb{N}} b_n$ is the unique element in $\bigcap_{n \in \mathbb{N}} B_n \subset F^*$.

Such map is injective: if $x_1 < x_2$ are distinct point in $[0, 1)$, then there exists n such that $\frac{1}{2^n} \leq \frac{1}{n} < x_2 - x_1$. Then $x_1 \in A_{nk}$ and $x_2 \in A_{nl}$ where $n \neq l$, hence they are mapped to disjoint half-open intervals B_1 and B_2 in F_n^* . Now $\mathcal{F}(x_1) \in B_1$ and $\mathcal{F}(x_2) \in B_2$, so they must not be equal.

Such map is surjective: if $y \in F^*$, then for any $n \in \mathbb{N}$, $y \in B_n \subset F_n^*$ where B_n is a half-open interval in the construction of F_n^* . $\mathcal{F}^{-1}$ then maps B_n to some

A_n . Since $B_{n+1} \subset B_n$, $A_{n+1} \subset A_n$. Let a_n be the endpoint of A_n , and b_n the left endpoint of B_n . Now $x = \sup_{n \in \mathbb{N}} a_n$ is the unique element in $\bigcap_{n \in \mathbb{N}} A_n$. By definition $\mathcal{F}(x) = \sup_{n \in \mathbb{N}} b_n$. This is the unique element in $\bigcap_{n \in \mathbb{N}} B_n$. Since $y \in \bigcap_{n \in \mathbb{N}} B_n$, $\mathcal{F}(x) = y$. $\square$

6.M, 7.F, 7.G, 7.J, 7.L, 3.F, 3.J

1. 6.M. Modify the argument given in Theorem 6.8 to show that if $a > 0$, then the number

$$b = \sup \{y \in \mathbf{R} : 0 \leq y, \quad y^2 \leq a\}$$

exists and has the property that $b^2 = a$. This number will be denoted by $\sqrt{a}$ or $a^{1/2}$, and is called the positive square root of a .

Proof. Denote the set by S . We first show that the supremum exists. Clearly $0 \in S$, so the set is nonempty. Next we show that the set is bounded from above. We split into cases where $x \leq 1$ and $x > 1$. When $x \leq 1$, we claim that the set is bounded above by 1. Suppose not, then there exists some $s \in S$ such that $s > 1$, hence $s^2 > 1$ but we have $s^2 \leq a \leq 1$, a contradiction. When $x > 1$, we claim that the set is bounded above by a . Suppose not, then there exists some $s \in S$ such that $s > a$, hence $s^2 > a^2$ but we have $s^2 \leq a < a^2$ (the second inequality is due to $a > 1$), a contradiction. In both cases the set is bounded above. By supremum property of $\mathbb{R}$, $b = \sup S$ exists.

Now we show $b^2 = a$. Suppose not, then by trichotomy property of real ordering, we have either $b^2 < a$ or $b^2 > a$. When $b^2 < a$, we take $n \in \mathbb{N}$, such that $\frac{1}{n} < \frac{a-b^2}{2b+1}$ (using Corollary 6.7 (b) of Archimedean Property). In this case

$$(b + \frac{1}{n})^2 = b^2 + \frac{1}{n} \left(2b + \frac{1}{n}\right) \leq b^2 + \frac{1}{n} (2b + 1) < b^2 + (a - b^2) = a.$$

This means $b + \frac{1}{n} \in S$, contrary to the fact that b is an upper bound of S .

When $b^2 > a$, we take $n \in \mathbb{N}$ such that $\frac{1}{n} < \frac{b^2-a}{2b}$ (such n exists by Corollary 6.7(b) of the Archimedean Property). Since $b = \sup S$, there exists an $s_0 \in S$ such that $b - \frac{1}{n} < s_0$. But this implies that

$$a < b^2 - 2\frac{b}{n} < b^2 - 2\frac{b}{n} + \frac{1}{n^2} = \left(b - \frac{1}{n}\right)^2 < s_0^2.$$

But this means $s_0^2 > a$, contrary to the fact that $s_0 \in S$.

We have excluded the possibilities of $b^2 < a$ and $b^2 > a$, so the only possibility left is $b^2 = a$. $\square$

2. 7.F. Let $J_n = (0, 1/n)$ for $n \in \mathbb{N}$. Show that this sequence of intervals is nested, but that there is no common point.

Proof. We first show the sequence is nested. Fix some $n \in \mathbb{N}$. It suffices to show that $J_{n+1} \subset J_n$. Take some $x \in J_{n+1}$, then by definition we have

$0 < x < \frac{1}{n+1}$. But since $\frac{1}{n+1} < \frac{1}{n}$, we have $0 < x < \frac{1}{n}$, hence by definition again we have $x \in J_n$.

Now we show there is no common point. Fix any $x \in J_1 = (0, 1)$. By Corollary 6.7(b), there exists some $n \in \mathbb{N}$ such that $0 < \frac{1}{n} < x$. In other words, $x \notin J_n$. Therefore, $\bigcap_{n \in \mathbb{N}} J_n = \emptyset$ since each element is not in at least one of the sets. $\square$

3. 7.G. If $I_n = [a_n, b_n]$, $n \in \mathbb{N}$, is a nested sequence of closed cells, show that

$$a_1 \leq a_2 \leq \cdots \leq a_n \leq \cdots \leq b_m \leq \cdots \leq b_2 \leq b_1.$$

If we put $\xi = \sup \{a_n : n \in \mathbb{N}\}$ and $\eta = \inf \{b_m : m \in \mathbb{N}\}$, show that $[\xi, \eta] = \bigcap_{n \in \mathbb{N}} I_n$.

Proof. Take $n, m \in \mathbb{N}$ where $n \leq m$. By assumption $I_m \subset I_n$. We need to show that $a_n \leq a_m$ and $b_m \leq b_n$. By definition of closed intervals, for all $x \in \mathbb{R}$, $x \in I_n \Leftrightarrow a_n \leq x \leq b_n$ and $x \in I_m \Leftrightarrow a_m \leq x \leq b_m$. By set inclusion, $x \in I_m \Rightarrow x \in I_n$. Therefore

$$a_m \leq x \leq b_m \Rightarrow a_n \leq x \leq b_n.$$

Suppose that $a_m < a_n$. Then there exists some x such that $a_m < x < a_n$, such that $a_n \leq x \leq b_n$ is not true, therefore the implication is not true for the specific x , a contradiction to the universally quantified statement. By trichotomy of real order, we have $a_m \geq a_n$. By the same reasoning we have $b_m \leq b_n$.

Now we show that $[\xi, \eta] = \bigcap_{n \in \mathbb{N}} I_n$. We first show that $[\xi, \eta] \subset \bigcap_{n \in \mathbb{N}} I_n$. Take $x \in [\xi, \eta]$, and take an arbitrary $n \in \mathbb{N}$. It suffices to show that $x \in I_n$. Since ξ is a supremum, we must have $\xi \geq a_n$. Therefore $x \geq \xi \geq a_n$. Similarly we have $x \leq b_n$. By definition of I_n , we have $x \in I_n$.

Next we show that $[\xi, \eta] \supset \bigcap_{n \in \mathbb{N}} I_n$. Take $x \in \bigcap_{n \in \mathbb{N}} I_n$. Then $x \in I_n$ for all $n \in \mathbb{N}$. Suppose that $x < \xi$. Then there exists some $m \in \mathbb{N}$ such that $x < a_m \leq \xi$. Clearly $x \in I_m$ is not true, a contradiction. Therefore we must have $x \geq \xi$. By the same reasoning we have $x \leq \eta$, so $x \in [\xi, \eta]$. $\square$

4. 7.J. Every open interval (a, b) which contains a point of $\mathbf{F}$ also contains an entire "middle third" set which belongs to $\mathcal{C}(\mathbf{F})$. Hence $\mathbf{F}$ does not contain any non-void open interval.

Proof. Fix some interval $(a, b) \subset \mathbb{R}$ and some point $x \in \mathbf{F} \cap (a, b)$. Since $a < x < b$, we have $b - x > 0$ and $x - a > 0$. We can choose some $n \in \mathbb{N}$ such that

$$3^n > \max\left(\frac{1}{x-a}, \frac{1}{b-x}\right).$$

To justify this, note that by Bernoulli's Inequality

$$3^n = (1 + 2)^n \geq 1 + 2n > n$$

and by Archimedean property we can choose n such that $n > \max\left(\frac{1}{x-a}, \frac{1}{b-x}\right)$.

By Corollary 6.7(c) of the Archimedean property, there exists $k \in \mathbb{N}$ such that $k \leq 3^n x < k + 1$. Since $3^n x - 3^n a > 1$, we have $k > 3^n x - 1 >$

3^na . Since $3^nb - 3^nx > 1$, we have $k + 1 \leq 3^nx + 1 < 3^nb$. Therefore $\left[\frac{k}{3^n}, \frac{k+1}{3^n}\right] \subset (a, b)$.

By definition and construction, $x \in \left[\frac{k}{3^n}, \frac{k+1}{3^n}\right] \subset F_n$. F_{n+1} takes the middle third from this interval, which is $\left[\frac{3k}{3^{n+1}}, \frac{3k+1}{3^{n+1}}\right] \cup \left[\frac{3k+2}{3^{n+1}}, \frac{3k+3}{3^{n+1}}\right] \subset F_{n+1}$. The middle-third $M = \left(\frac{3k+1}{3^{n+1}}, \frac{3k+2}{3^{n+1}}\right)$ satisfies the properties $M \subset F_n$ and $M \subset (a, b)$. $\square$

5. 7.L. Show that $\mathbf{F}$ is not the union of a countable collection of closed intervals.

Proof. Suppose $\mathbf{F} = \bigcup_{n \in \mathbb{N}} I_n$ where each $I_n = [a_n, b_n]$. Suppose there exists some $m \in \mathbb{N}$ such that $a_m < b_m$. Then the open interval (a_m, b_m) is nonempty, and has a nonempty intersection with $\mathbf{F}$. By Exercise 7.J, the interval (a_m, b_m) contains a "middle-third" set. This contradicts the fact that $[a_n, b_n] \subset \mathbf{F}$.

Therefore $a_n \geq b_n$ for all $n \in \mathbb{N}$. If $a_n > b_n$ then the interval is empty, so we have $I_n = \{x_n\} \subset \mathbf{F}$ for at most countable number of I_n , and the rest of them are empty. This is to say that the set $\mathbf{F}$ is at most countable. But we have concluded in Exercise 7.I that $\mathbf{F}$ is not countable. $\square$

6. 3.F. Use the fact that every infinite set has a denumerable subset to show that every infinite set can be put into one-one correspondence with a proper subset of itself.

Proof. Consider an infinite set X and a denumerable subset $A \subset X$. Denumerate A by $a_1, a_2, \dots, a_i, \dots$, and define the map

$$F : X \longrightarrow X \setminus \{a_1\}$$

$$x \longmapsto F(x) = \begin{cases} x & x \notin A \\ a_{i+1} & x = a_i \in A \end{cases}.$$

F is a function because for every x either $x \notin A$, or $x \in A$, in this case there must exist $i \in \mathbb{N}$ such that $x = a_i$. F is injective since for any $x_1 \neq x_2$, either they both $\notin A$, in this case $F(x_1) = x_1 \neq x_2 = F(x_2)$; or they are both $\in A$, in this case $F(x_1) = F(a_i) = a_{i+1} \neq a_{j+1} = F(a_j) = F(x_2)$; or one of them is in A and the other is not, in this case the claim is obvious.

F is surjective because for any $x \in X \setminus A$, its preimage is itself; and for any $x = a_i \in A \setminus \{a_1\}$ where $i \neq 1$, its preimage is $a_{i-1} \in A \setminus \{a_1\}$. $\square$

7. 3.J. Prove that $\mathbb{N}$ cannot be put into one-one correspondence with any initial segment of $\mathbb{N}$.

Proof. Suppose for the sake of contradiction that there exists some $n \in \mathbb{N}$ such that there is a bijection $f : \mathbb{N} \rightarrow S_n$. Hence in particular f is injective. Then by restricting the domain, $f|_{S_{n+1}} : S_{n+1} \rightarrow S_n$ is injective. However this cannot be true since by the pigeonhole principle there must exist two elements in S_{n+1} that are mapped to the same same element in S_n . $\square$

Homework 3: Week 3 due Sep 16

Fri 09 Sep 2022 16:50

8.F, 8.G, 8.M, 8.Q, 9.G, 9.H, 9.L, 9.Q(a,b,c,d)